

A Theory of Economic Slack

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Terminology

Before we start, we define the core terms used throughout the book. Some of the terms are not used commonly in economics, some are often misconstrued, and some are used in the book in a way that is atypical. Either way, it is useful to clarify the meaning of these terms upfront.

The term *slack* describes situations in which goods or services are available for sale but trades do not occur. Sometimes the goods are already produced; sometimes they are available to be produced at negligible marginal cost. In both cases, trades that would generate value for sellers and buyers do not materialize.

The term *slackish* describes a class of markets where slack is the central mechanism that equalizes supply and demand. Slackish does not mean the market is always too slack; rather, it refers to the structural mechanism through which the market operates. It means that market tightness is the key mediating variable, whether the market is inefficiently slack or inefficiently tight.

The term *equilibrium* is used to denote time-invariant points of differential equations or dynamical systems (instead of *steady state*). An equilibrium point has $\dot{x}(t) = 0$. This follows usage in dynamical systems and conveys the idea of a balanced state, which may be reached quickly.

The term *solution* is used to denote the allocation that satisfies all the equations of a model (instead of *equilibrium*). This describes more transparently what the allocation is.

The term *matching function* refers to the aggregate function that mediates trades in slackish markets. With s sellers and b buyers looking to trade, the matching function $m(s, b)$ gives the number of trades that take place. It summarizes the random, costly, imperfect process of finding trading partners without modeling that process in detail.

The term *tightness* refers to the central equilibrating variable in slackish markets. Tightness is denoted by θ and defined as the ratio of the two arguments in the matching function: the ratio b/s between the number of buyers and the number of sellers. Since trading probabilities are determined by tightness, tightness summarizes the state of the

market.

The term *matching elasticity* refers to the elasticity of the matching function with respect to the number of sellers, holding the number of buyers fixed. The matching elasticity is a structural property of the matching function, denoted by η , and generally endogenous to tightness.

The term *matching wedge* denotes the gap between goods purchased and goods consumed caused by matching costs. Buyers must devote part of their expenditure to matching costs, so the amount spent on goods and services that are actually consumed is less than the total amount spent. If the wedge is τ , consuming one unit requires purchasing $1 + \tau$ units. Because matching probabilities depend on tightness, the wedge is endogenous to tightness. In the labor market, where matching costs take the form of recruiting costs, we refer to the matching wedge as the *recruiting wedge*.

The term *market supply* denotes the amount of goods actually sold given market tightness and market price. It is the effective supply; the *market capacity* is the notional market supply: the amount of goods that sellers place on the market and would like to sell given market tightness and market price. Because each good sells only with probability less than one, effective supply falls short of notional supply.

The term *market demand* denotes the amount of goods that buyers actually purchase given market price and market tightness. That is the effective demand. The notional market demand is the amount that buyers would like to consume. Because consuming one good requires purchasing more than one, effective demand exceeds notional demand. Buyers purchase goods from sellers through *visits*. Visits are the demand-side input into the market's matching function, just as goods for sale are the supply-side input.

The term *unemployed* describes workers who are available and willing to work but unable to find a job. This definition is consistent with that used by the US Bureau of Labor Statistics (BLS). Based on the definition, the concept of voluntary unemployment is an oxymoron. Being unemployed means wanting a job but not finding one: it is by definition involuntary. Someone who does not have a job and does not want one (which is presumably what is meant with voluntary unemployment) is *out of the labor force*—not unemployed. Unemployed workers correspond to slack on the labor market.

The term *job vacancies* denotes positions that firms report as open and for which they are recruiting. This definition is consistent with that used by the BLS. Vacancies measure recruiting effort; they are not a measure of unmet labor demand. Some vacancies never result in hiring, because the job-filling probability is always less than 1. Job vacancies correspond to visits on the labor market.

The term *full employment* describes a labor market that operates efficiently. Accordingly, the full-employment rate of unemployment (FERU) is the socially efficient rate of unemployment. Full employment therefore does not mean zero unemployment and does

not automatically imply stable prices.

The term *unemployment gap* denotes the distance of the labor market from social efficiency. It is computed by comparing the actual unemployment rate to the FERU.

The term *Beveridge curve* denotes the negative relationship between the slack rate and the visit rate. In the labor market specifically, it is the negative relationship between the unemployment rate and the vacancy rate. In dynamic slackish models, it appears once market flows are balanced and the slack rate reaches equilibrium.

The term *Okun curve* denotes the relationship between the logarithm of output and slack (traditionally *Okun's law*). The labor-market Okun curve is the relationship between log output and labor-market slack—that is, unemployment. The product-market Okun curve is the relationship between log output and product-market slack. The slopes of the Okun curves are used to identify the shocks that drive business cycles.

The term *Phillips curve* denotes the negative relationship between inflation and slack. The slackish Phillips curve links inflation to the unemployment gap, unlike the New Keynesian Phillips curve, which links inflation to an output gap or marginal costs.

The term *sufficient statistics* refers to statistics used in formulas describing efficient allocations and optimal policies. They have two defining properties, which distinguish them from structural parameters. First, they should be valid across a broad range of models. Second, they should be estimable directly from data.

The terms *positive* and *negative* are used for $x \geq 0$ and $x \leq 0$ (instead of *nonnegative* and *nonpositive*). The terms *strictly positive* and *strictly negative* are used for $x > 0$ and $x < 0$. This follows French usage and avoids the cognitive load from “non-” constructions.

Similarly, the terms *increasing* and *decreasing* are used for weakly monotone functions (instead of *nondecreasing* and *nonincreasing*). The terms *strictly increasing* and *strictly decreasing* are used for strictly monotone functions. This again avoids the cognitive load from “non-” constructions.